

backend/core/ollama_manager.py Documentation (Roman Urdu)

Yeh file project ko properly "Solo" / standalone banati hai kiyun ke **user ko pehlay se koi Ollama application download karke install nahi karni parti**. Yeh Backend zud detect karta hai aur khod hi background mein sab files utha leta hai.

File Ka Maqsad (Purpose)

Ollama binary (exe files) ka Lifecycle manager. Yeh net se download karnay, ZIP extract karne, aur binary ko aik invisible sub-process me Start / Stop karne ke liye zimmdar hai bina user ko pareshan kiye.

Code Blocks aur Functions ki Tafseel

1. AVAILABLE_MODELS_CATALOG

- **Kya hai?:** Yeh aik bari static list hai code me, jisme tamam ahem Models (Llama 3.2, Qwen 2.5, Mistral, TinyLlama) aur unke RAM ya VRAM ki demand likhi gai hai. Issi list ko Frontend apne Model Downloader Catalog (market) mein dikhata hai.

2. class ollamaManager

- **Tafseel:** Sub process handle karnay wali fully async main body.
- **ensure_running():** Sub se pehlay API request pe 11434 (port) ping krti ha check karny k agar Ollama pehley sr running hai toh done karo. Warna check krti h kya data/ollama/ollama.exe hai? Agar ha tou `_start()` kro, warana `_download_binary()` chalao.
- **_download_binary() / _extract_zip():** Internet (HTTPS) hit kar ke Ollama server se Linux (.tgz) ya windows (.zip) download krti ha, extract krti ha aur progress UI ko percent (0 to 100) deti ha.
- **_start():** Extract hone k bad automatically `subprocess.Popen` se executable chalata hai environment variable (`OLLAMA_MODELS`) setup karkar k models desktop ki c drive C:\Users ni bulk folder /data/models m store hn taa k portability remain kary. Max power enable ho tou wo override parameters b pass krrata ha!

- `get_model_catalog(vram_mb, ram_mb)`: User ka PC specifications hardware.py se mangwa kr automatically recommend kr deta h k is catalog / market m sy konsa AI system fit bethega apk liy ("Tight fit", "Runs on GPU" wagera tags assign kr deta ha).